

## Detailed Experimental Results and Implementation Details

### A. Summary of Quantitative Results

To provide a more rigorous assessment of the evaluated vision-language models (VLMs), we present the complete numeric performance across three downstream tasks: Zero-shot Classification, Semantic Segmentation, and Visual Question Answering (VQA). All reported values are averaged over three independent runs where applicable, with standard deviations provided in parentheses to ensure statistical significance.

Table 1 compares the performance of CLIP variants and SigLIP2 across clean (ImageNet-1K) and natural adversarial datasets.

**Table 1: Zero-shot Classification Top-1 Accuracy (%)**

| Model Variant          | ImageNet-1k                        | IN-A                               | IN-Typo                            | RTA-100                            | LangAdv                            |
|------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| <b>OpenAI CLIP</b>     |                                    |                                    |                                    |                                    |                                    |
| <b>ViT-B/16</b>        | 63.9 ( $\pm 0.1$ )                 | 47.6 ( $\pm 0.2$ )                 | 48.9 ( $\pm 0.4$ )                 | 42.4 ( $\pm 0.3$ )                 | 32.5 ( $\pm 0.1$ )                 |
| <b>ViT-B/32</b>        | 59.3 ( $\pm 0.1$ )                 | 60.8 ( $\pm 0.1$ )                 | 49.1 ( $\pm 0.3$ )                 | 46.6 ( $\pm 0.3$ )                 | 38.0 ( $\pm 0.1$ )                 |
| <b>ViT-L/14</b>        | 70.9 ( $\pm 0.1$ )                 | 68.0 ( $\pm 0.2$ )                 | 49.8 ( $\pm 0.3$ )                 | 42.0 ( $\pm 0.2$ )                 | 37.5 ( $\pm 0.1$ )                 |
| <b>ResNet50</b>        | 54.0 ( $\pm 0.2$ )                 | 22.5 ( $\pm 0.1$ )                 | 36.5 ( $\pm 0.2$ )                 | 27.1 ( $\pm 0.2$ )                 | 20.1 ( $\pm 0.2$ )                 |
| <b>Robust CLIP</b>     |                                    |                                    |                                    |                                    |                                    |
| <b>ViT-B/16-fare4</b>  | 54.1 ( $\pm 0.2$ )                 | 10.2 ( $\pm 0.2$ )                 | 48.5 ( $\pm 0.4$ )                 | 41.4 ( $\pm 0.2$ )                 | 35.2 ( $\pm 0.3$ )                 |
| <b>ViT-B/16-tecoa4</b> | 59.3 ( $\pm 0.2$ )                 | 8.2 ( $\pm 0.2$ )                  | 57.1 ( $\pm 0.3$ )                 | 42.1 ( $\pm 0.3$ )                 | 39.5 ( $\pm 0.2$ )                 |
| <b>ViT-B/32-fare4</b>  | 49.3 ( $\pm 0.3$ )                 | 6.7 ( $\pm 0.2$ )                  | 45.4 ( $\pm 0.5$ )                 | 45.0 ( $\pm 0.3$ )                 | 35.1 ( $\pm 0.1$ )                 |
| <b>ViT-B/32-tecoa4</b> | 54.0 ( $\pm 0.3$ )                 | 5.2 ( $\pm 0.2$ )                  | 51.9 ( $\pm 0.4$ )                 | 40.3 ( $\pm 0.2$ )                 | 34.6 ( $\pm 0.3$ )                 |
| <b>ViT-L/14-fare2</b>  | 68.7 ( $\pm 0.2$ )                 | 34.2 ( $\pm 0.3$ )                 | 51.7 ( $\pm 0.3$ )                 | 31.6 ( $\pm 0.3$ )                 | 30.1 ( $\pm 0.2$ )                 |
| <b>ViT-L/14-tecoa2</b> | 72.7 ( $\pm 0.3$ )                 | 29.1 ( $\pm 0.2$ )                 | 68.2 ( $\pm 0.3$ )                 | 46.1 ( $\pm 0.3$ )                 | 33.2 ( $\pm 0.3$ )                 |
| <b>ViT-L/14-fare4</b>  | 64.2 ( $\pm 0.2$ )                 | 19.0 ( $\pm 0.3$ )                 | 51.1 ( $\pm 0.3$ )                 | 34.9 ( $\pm 0.2$ )                 | 36.0 ( $\pm 0.1$ )                 |
| <b>ViT-L/14-tecoa4</b> | 66.6 ( $\pm 0.3$ )                 | 18.4 ( $\pm 0.3$ )                 | 62.4 ( $\pm 0.4$ )                 | 42.2 ( $\pm 0.4$ )                 | 37.8 ( $\pm 0.4$ )                 |
| <b>SigLIP2</b>         |                                    |                                    |                                    |                                    |                                    |
| <b>B/16-224</b>        | 68.7 ( $\pm 0.1$ )                 | 53.7 ( $\pm 0.3$ )                 | 64.5 ( $\pm 0.2$ )                 | 72.8 ( $\pm 0.3$ )                 | 52.0 ( $\pm 0.2$ )                 |
| <b>B/16-256</b>        | 69.7 ( $\pm 0.1$ )                 | 58.2 ( $\pm 0.2$ )                 | 65.4 ( $\pm 0.1$ )                 | 74.6 ( $\pm 0.2$ )                 | 54.0 ( $\pm 0.3$ )                 |
| <b>B/16-384</b>        | 71.1 ( $\pm 0.1$ )                 | 70.5 ( $\pm 0.2$ )                 | 67.1 ( $\pm 0.2$ )                 | 76.3 ( $\pm 0.3$ )                 | 56.0 ( $\pm 0.2$ )                 |
| <b>B/16-512</b>        | 71.4 ( $\pm 0.1$ )                 | 75.0 ( $\pm 0.2$ )                 | 67.3 ( $\pm 0.2$ )                 | 78.0 ( $\pm 0.4$ )                 | 56.2 ( $\pm 0.2$ )                 |
| <b>B/32-256</b>        | 65.4 ( $\pm 0.1$ )                 | 36.3 ( $\pm 0.3$ )                 | 61.4 ( $\pm 0.3$ )                 | 70.5 ( $\pm 0.3$ )                 | 50.5 ( $\pm 0.2$ )                 |
| <b>L/16-256</b>        | 71.9 ( $\pm 0.2$ )                 | 70.3 ( $\pm 0.2$ )                 | 65.2 ( $\pm 0.3$ )                 | 80.0 ( $\pm 0.3$ )                 | 58.5 ( $\pm 0.3$ )                 |
| <b>L/16-512</b>        | <b>72.8 (<math>\pm 0.2</math>)</b> | <b>81.8 (<math>\pm 0.3</math>)</b> | <b>72.8 (<math>\pm 0.3</math>)</b> | <b>80.7 (<math>\pm 0.2</math>)</b> | <b>61.0 (<math>\pm 0.4</math>)</b> |